

# Supporting Information

## For

Graph Symmetry and Kekulé Structure Localization Correlate with Carcinogenic Potency of Polycyclic Aromatic Hydrocarbons

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## Contents

- Table S1. PubChem confirmations and graph-verification notes for corrected or high-risk structures (extended commentary on five cases requiring explicit annotation).
- Table S2. Complete 27-PAH descriptor table used in the main analysis. The `CID` column lists the canonical PubChem identifier for every entry; all 27 structures were programmatically retrieved from PubChem and the resulting molecular graphs were independently verified before being passed to the  $K$  and  $|Aut(G)|$  computations.
- Table S3. Sensitivity analyses for floor assignment and benzo[*c*]phenanthrene proxy assignment.
- Table S4. Leave-one-out predictions and residuals for every molecule. The three methylated PAHs (3-methylcholanthrene, 5-methylchrysene, and 7,12-dimethylbenz[*a*]anthracene) — the systematic failure mode discussed in the main text — appear with LOO residuals of +1.30, +1.78, and +2.41  $\log_{10}$  units respectively, all in the direction of underprediction; this is the basis for the decomposition  $RMSE_{unsubstituted} = 1.15$  vs  $RMSE_{methylated} = 1.88$ .
- Table S5. Dibenzopyrene isomer comparison, including the qualitative comparator dibenzo[*e,l*]pyrene.
- Figure S1. ROC curve for the binary classification task at threshold  $K/|Aut(G)| \geq 5.0$  across the full 27-PAH dataset (full-set AUC = 0.95; EPA Priority subset  $n=14$  AUC = 0.95). Exported as `figS1_ROC.pdf` and `figS1_ROC.png`.

## Selection of the binary threshold $K/|Aut(G)| \geq 5.0$

The threshold was selected at the natural gap in the descriptor distribution (Table S2): dibenz[*a,h*]anthracene ( $K/|Aut| = 5.00$ , PEF = 5.0) is the lowest- $K/|Aut|$  compound classified as a regulatory carcinogen in the primary hierarchy and serves as the anchor point. Shifting the cutoff by  $\pm 0.5$  around 5.0 trades a single boundary case in each direction without improving the overall confusion matrix; the value 5.0 was therefore retained as the simplest defensible cut. The two false positives (benzo[*ghi*]perylene, benzo[*e*]pyrene) and two false negatives

(benzo[k]fluoranthene, dibenz[a,h]anthracene) are explicitly retained and discussed in the main text rather than tuned away.

## Key SI Notes

- All 27 structures are cross-verified against PubChem entries (Table S2, CID column). The five cases below required explicit annotation because they involved nomenclature ambiguity, prior literature errors, or boundary failure modes; the remaining 22 structures matched their canonical PubChem entries on the first retrieval.
- 5-Methylchrysene is confirmed against PubChem CID 19427.
- 7,12-Dimethylbenz[a]anthracene (DMBA) is confirmed against PubChem CID 6001.
- Benzo[k]fluoranthene remains a real boundary false negative after pynauty confirmation of  $|Aut| = 2$ .
- Dibenz[a,e]pyrene is corrected to PubChem CID 9126 with  $K = 17$  and  $|Aut| = 1$ .
- Dibenz[e,l]pyrene maps to PubChem CID 9122 and gives  $K = 20$ ,  $|Aut| = 4$ ,  $K/|Aut| = 5.0$ .

## Exported Files

- [Table S1](#)
- [Table S2](#)
- [Table S3](#)
- [Table S4](#)
- [Table S5](#)
- [Figure S1 PDF](#)
- [Figure S1 PNG](#)

## LOO Summary

- LOO Spearman rho = +0.708 (95% bootstrap CI: +0.480 to +0.834)
- LOO RMSE = 1.25 log10 units
- RMSE unsubstituted (n=24) = 1.15
- RMSE methylated (n=3) = 1.88
- LOO classification accuracy = 19/27 (70.4%)

## Binary Summary

- Threshold  $K/|Aut| \geq 5.0$
- TP = 11, TN = 12, FP = 2, FN = 2 (sum = 27)
- AUC = 0.95 (full dataset); AUC = 0.95 (EPA Priority PAHs only, n=14)

## Bootstrap 95% CI

- Primary  $\rho$  = +0.745 (95% CI: +0.546 to +0.855, n=10000 resamples, seed=42)
- EPA-only (n=14):  $\rho$  = +0.759,  $p$  = 0.0016, AUC = 0.95